# Administrator Manual — Tracker Porti

> 🇮🇹 Manuale in italiano: manuale_admin.md · index.html

Guide to the functions reserved for Tracker Porti **administrators**: user and group management, coverage-map controls, risk model, system logs, and editing configuration files.

> This manual **complements** the User manual, which covers day-to-day use of the app. Here you'll find **only** what is reserved for administrators. For common features (monitoring, followed ships, ship detail, notifications, export…) refer to the user manual.

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## Administrator role

Administrators see the **Admin** link at the top right, which opens the **administration page** ( `/admin` ). An administrator can:

- **approve** pending new registrations;
- **enable or disable** an account;
- **change a user's role** (regular user ↔ administrator ↔ tester);
- **reset a user's password** — generates a **one-time link** (valid 24 hours) to hand to them;
- **delete** a user;
- **create and manage user groups** (see below);
- **impersonate** a user — view their areas, monitoring, and followed ships in **read-only** mode, with a prominent banner and one-click exit;
- consult the **logs** (activity log and API log), shared and visible only to administrators.

Settings managed by administrators (data sources, sanctions/PSC screening, risk-score weights) are **global** for all users. In the Settings interface, the tabs and toggles reserved for administrators are **hidden** from regular users (and protected server-side regardless).

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# User groups

An administrator can put several users into a **group**. When a user is part of a group, they **share with the other members** (as a **union** of what each one had):

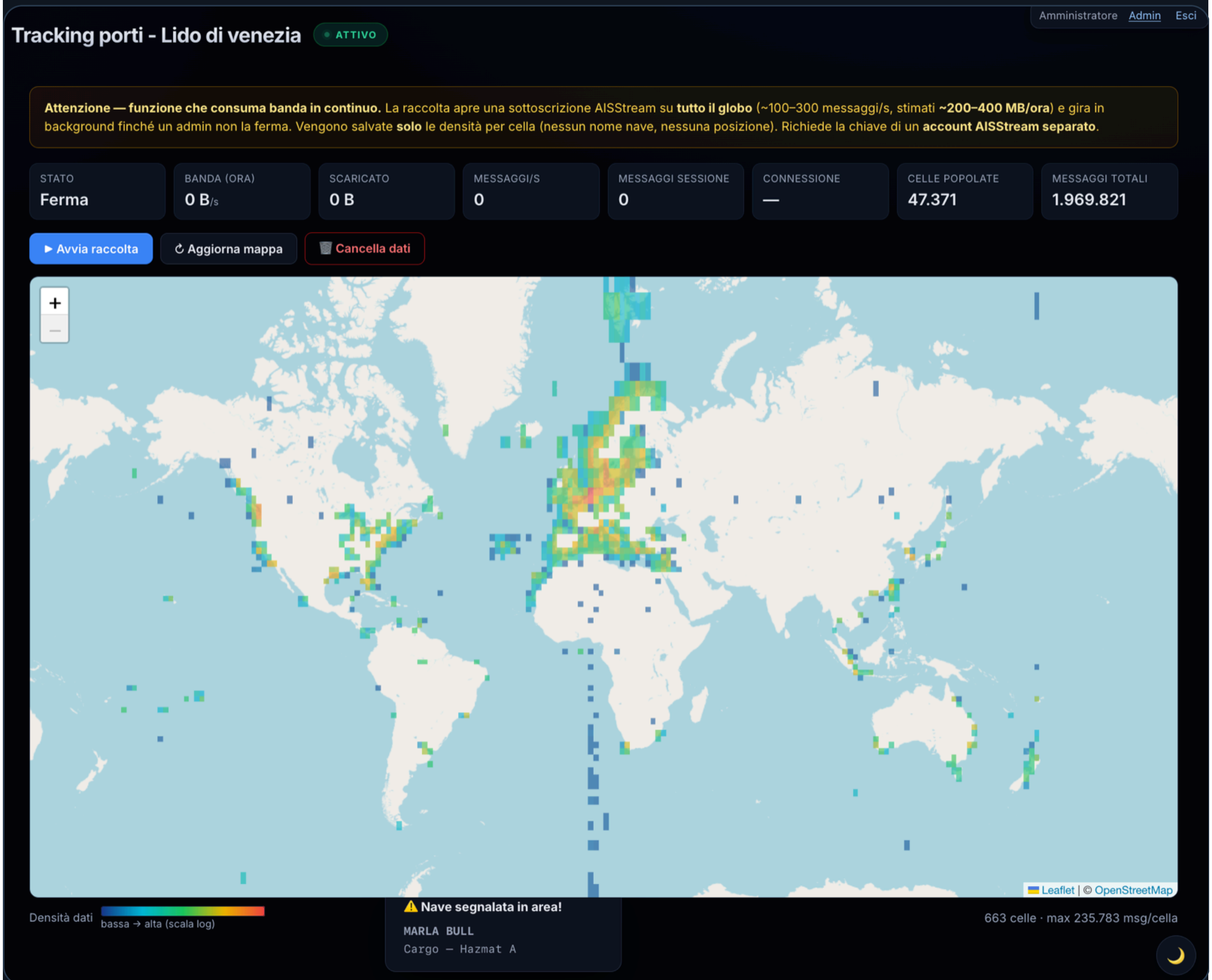
- monitoring **areas**;
- **followed ships**, **flagged ships ★**, and **muted ships 🔕**;
- **notification preferences** (app and Telegram) and **map display**, plus the **default area**.

In practice: if one member adds an area, follows a ship, or enables a notification, **the other members get it** on their next visit — and vice versa (write-through sync). What stays **personal** to each user: the **Telegram link** (their own chat) and the interface **language**.

**Management (from the Admin page):** a group has a **name**, a **description**, and **at least 2 members**. On creation you choose the **template user** whose settings the group starts from. You can **add/remove** members or **dissolve** the group.

> A removal that would leave **only 1 member** is blocked: in that case, **dissolve** the group instead. If an administrator removes a user from a group, that user **keeps** everything that was shared up to that point (areas, ships, settings): they simply stop syncing with the others.

# Coverage map — administrator controls




*World coverage map: grid colored by AIS message density.*

The Coverage map is visible read-only to all users (and even without login at `/heatmap`). Only **administrators** can additionally:

- **start** and **stop** data collection (**Start/Stop collection** buttons);
- see **real-time connection statistics** (bandwidth used, messages per second, populated cells...);
- **Refresh map** and **Clear data** (empties the collected grid).

**How collection works:** once started, it runs **in the background** on the server — even if no one has the page open — until an administrator stops it. It resumes on its own after a server restart. As a safety measure, it **automatically shuts off** if no user has been active for **10 minutes**.

> **Warning:** while collection is active, the app **downloads data from the whole world** continuously (roughly **200–400 MB per hour**). The feature requires an **AISStream key from a separate account** (different from the one used for regular monitoring), set in `local.properties` as `HEATMAP_AIS_API_KEY`. Without that key, the feature stays off.

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## Risk model (signal weights)

Besides the **cargo-type weights** (editable by everyone in Settings), administrators can adjust **how many points each risk signal is worth** (AIS blackout, spoofing/position jump, dwell, draught increase, sanctions, PSC, GFW events, rendezvous, etc.) from ⚙ **Settings → "⚖ Risk model" section**.

- A **grid** with one field per signal: change the values and press 💾 **Save weights**. **Immediate** effect, no restart needed. **Reset to default** restores factory values.
- Setting a weight to **0** disables that signal.
- You can save full configurations as **risk profiles** (*Risk profile* menu → *Save as…*) and recall them with *Apply* — useful for quickly switching between different setups (e.g. a more aggressive profile, or one tuned for areas with poor AIS coverage).

> **Detection thresholds** and signal **multipliers** are not in the UI: they stay in the `app.config.properties` file (`RISK_*` keys).

The Settings **General** tab, "⚖ Risk model" section, also has two toggles **reserved for administrators** (invisible to other users):

- **Exclude tankers** — doesn't assign the "ship type" score to tanker hulls (AIS code 80–89). Useful when monitoring weapons transport, which a tanker can't carry out: it only zeroes that one factor, not the whole score (a tanker can still land in the red band from other signals — sanctions, AIS dark, etc.). It's a **global** value shared by every user, and different from the notifications' ship-type filter (Settings → **Notifications**, personal to each user): that one only decides what reaches you as a notification, without touching the score.
- **Check position jump** / **Check AIS blackout** — include those signals in the score. Disable them in areas with poor AIS coverage, where sparse reports produce false positives (apparent jumps/gaps that aren't real).

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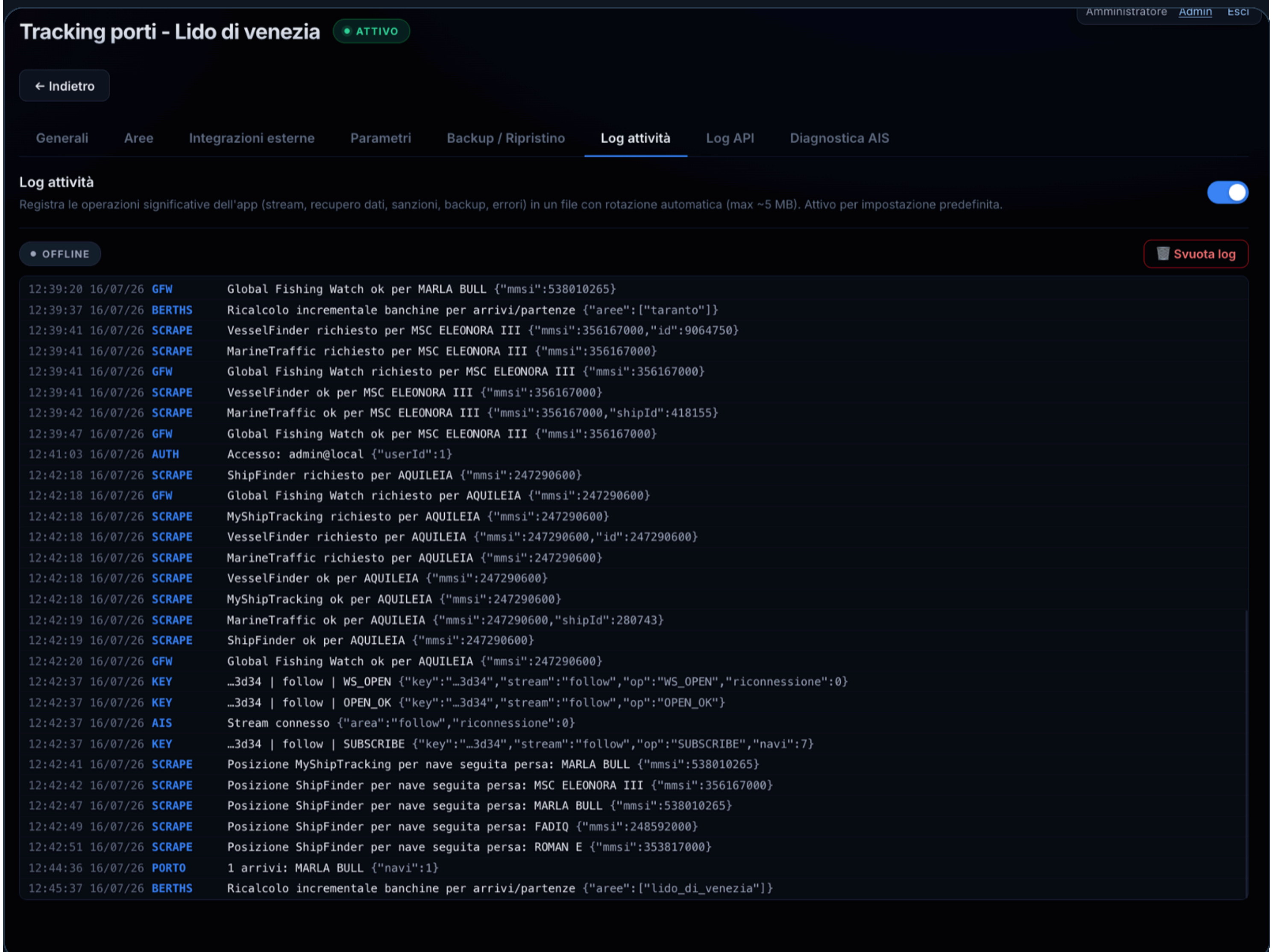
# System logs

Logs are **shared** and visible **only to administrators**, as tabs in Settings.

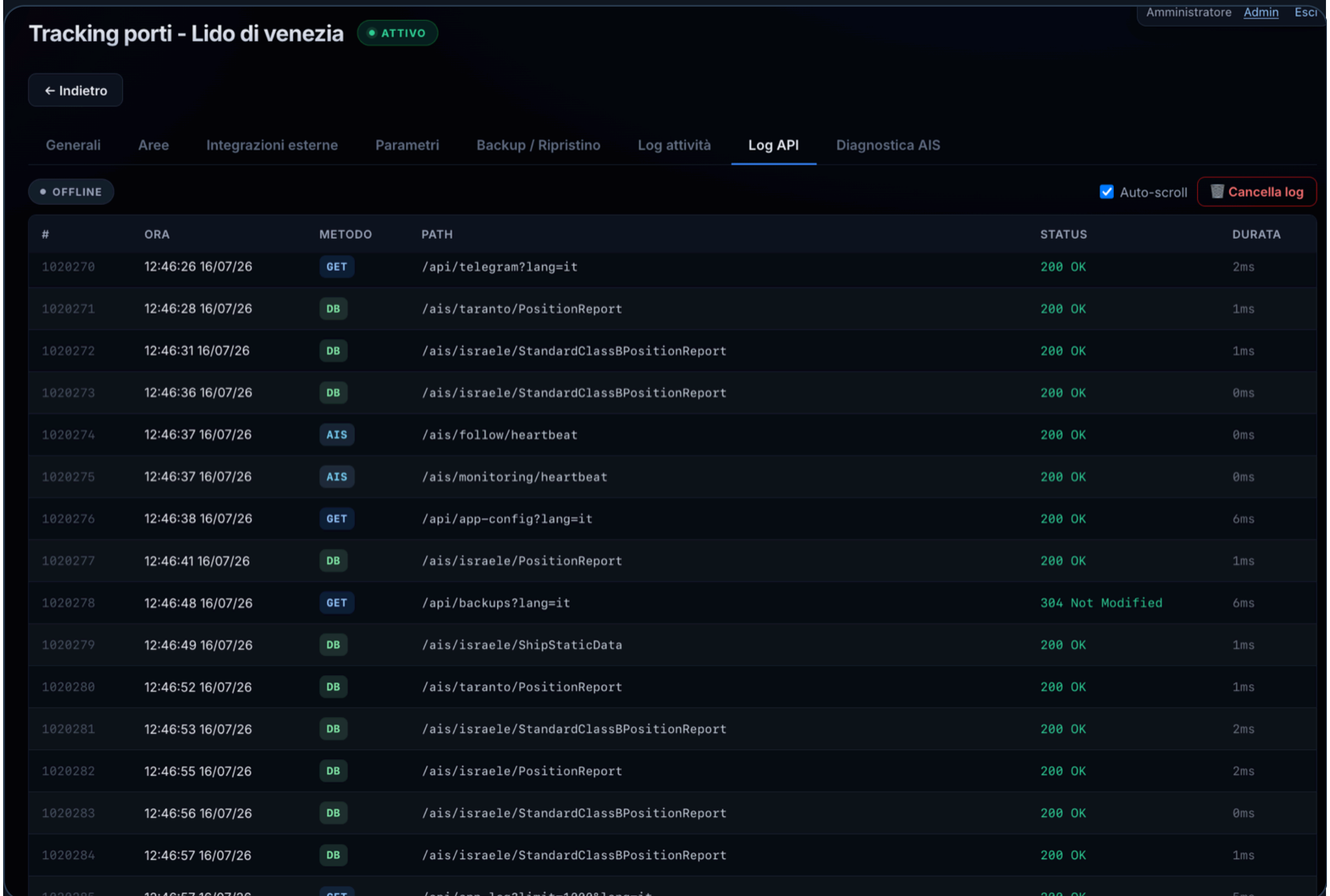
## Activity log




*Settings — Activity log: chronological record of the app's operations.*

Records the app's significant operations (streams, data fetching, sanctions, backups, errors) in a file with **automatic rotation** (max ~5 MB). On by default (**Activity log** toggle in the General tab, admin). The log can also be viewed from the 🪵 **Activity log** overlay in the sidebar and can be **cleared**.

## API log




*Settings — API log: list of /api calls with method, path, and outcome.*

Lists the application’s API calls in real time (method, path, status, time), useful for diagnostics. Sensitive request bodies (login, etc.) are **suppressed**: passwords are never logged.

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# Editing configuration files

Some advanced settings aren’t in the interface but in text files in the project folder. Open them with a text editor, change the values, and **restart the application** to apply them. Lines starting with `#` (or `//`) are comments and are ignored.

## `local.properties` — keys and secrets

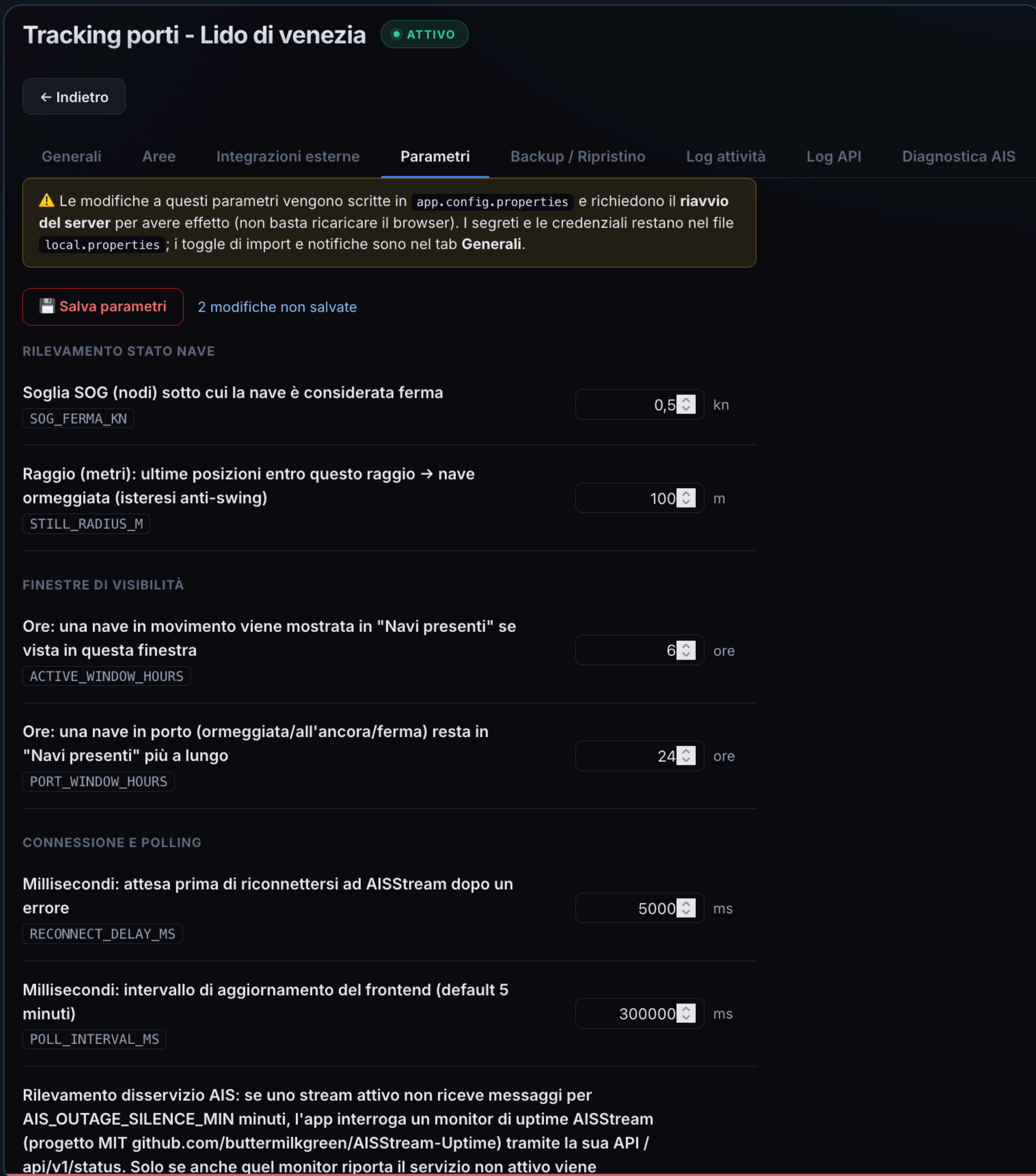
Contains the API key and initial preferences. Format `KEY=value`, one per line. **Must not be shared** (it contains the API key) and is in `.gitignore`. If it doesn't exist, copy it from `local.properties.example`.

| Key | Meaning |
| --- | --- |
| `AIS_API_KEY` | AISStream.io key (required) — used by the **monitoring areas'** streams |
| `FOLLOW_AIS_API_KEY` | AISStream.io key for the **followed ships** stream. Best from a **separate account** (see note). Empty = reuses `AIS_API_KEY` |
| `HEATMAP_AIS_API_KEY` | Key of a **separate** AISStream account, used **only** for the Coverage map. Empty = feature disabled. Must be written "bare", **with no comments on the same line** |
| `BBOX_PRESET` | Area shown at startup (an area's key, e.g. `bari`) |
| `IMPORT_VF_DATA` / `IMPORT_MT_DATA` | `true` / `false` — enable VesselFinder / MarineTraffic import |
| `IMPORT_SF_DATA` | `true` / `false` — ShipFinder import + position to re-locate lost followed ships |
| `IMPORT_MST_DATA` | `true` / `false` — MyShipTracking import (second, independent backup position source) |
| `IMPORT_SANCTIONS` | `true` / `false` — screening against the OFAC SDN list |
| `IMPORT_SANCTIONS_EXTRA` | `true` / `false` — EU / UK OFSI / UN lists (only with `IMPORT_SANCTIONS`); default `true` |
| `IMPORT_PSC` | `true` / `false` — Port State Control screening (Paris/Tokyo MoU flags + banned ships) |
| `IMPORT_EQUASIS` | `true` / `false` — on-demand Equasis lookup in the ship detail |
| `EQUASIS_USER` / `EQUASIS_PASSWORD` | Equasis account credentials (free registration at equasis.org) |
| `IMPORT_GFW` | `true` / `false` — Global Fishing Watch enrichment; **default `true`** |

| Key | Meaning |
| --- | --- |
| `GLOBAL_FISHING_WATCH_TOKEN` | Global Fishing Watch API (Bearer) token. Data free for non-commercial use only |
| `ADMIN_USERNAME` / `ADMIN_EMAIL` / `ADMIN_PASSWORD` | Default administrator (re-created at startup if missing). **Change the password** on a server reachable by others |
| `COOKIE_SECURE` | `true` / `false` — sends the session cookie only over HTTPS; set to `true` behind TLS |
| `SESSION_TTL_DAYS` | Login session length in days. Default `30` |

> 💡 **Recommended: three keys from three separate AISStream accounts.** AISStream limits connections **per account, not per key**. The app opens three independent streams — **area monitoring** (`AIS_API_KEY`), **followed ships** (`FOLLOW_AIS_API_KEY`), and **coverage map** (`HEATMAP_AIS_API_KEY`). If two use keys from the **same account**, they compete for the same slot and one keeps getting rejected. Give each feature a key from a **dedicated account**.

## `app.config.properties` — operating parameters




*Settings — Parameters: configuration fields grouped by category with description.*

Contains thresholds and parameters (time windows, radii, retention, berths, score weights). Format `KEY=value`, each parameter documented by a comment in the file. **You can also edit them from the interface** at ⚙ **Settings → Parameters** (more convenient). Either way, you need to **restart the server** to apply them (read only at startup). Examples:

| Key | Meaning | Default |
|---|---|---|
| `SOG_FERMA_KN` | Speed (knots) below which a ship is “stationary” | `0.5` |
| `ACTIVE_WINDOW_HOURS` | Hours a moving ship stays among the “current” ones | `6` |
| `PORT_WINDOW_HOURS` | Hours a ship in port stays among the “current” ones | `24` |
| `POLL_INTERVAL_MS` | Interface refresh interval (ms) | `300000` |
| `AIS_OUTAGE_CHECK` | Enables AIS outage detection | `true` |
| `AIS_OUTAGE_SILENCE_MIN` | Minutes without signals before querying the uptime monitor | `10` |
| `AIS_UPTIME_SELFHOST_URL` | URL of your own self-hosted uptime monitor instance (queried first) | *(empty)* |
| `AIS_UPTIME_URL` | URL of the public uptime monitor, used as a fallback | `https://aisuptime.buttermilkgreen.fyi` |
| `MAX_READINGS_PER_TYPE` | Maximum readings kept per message type | `10000` |
| `BERTH_CLUSTER_EPS_M` | Mooring-to-berth clustering radius (meters) | `80` |
| `BERTH_MIN_PTS` | Minimum nearby moorings to form a berth | `3` |

| Key | Meaning | Default |
|---|---|---|
| `BERTH_MIN_MOORINGS` | Minimum moorings before characterizing a berth | `10` |
| `BERTH_DOMINANT_PCT` | Percentage a category must exceed to name the berth | `60` |
| `BERTH_RECOMPUTE_MIN` | Minutes between one automatic berth recomputation and the next | `30` |
| `HEATMAP_GRID_DEG` | Coverage map cell size, in degrees (~28 km) | `0.25` |
| `HEATMAP_FLUSH_SEC` | How often (seconds) counts are written to disk | `10` |
| `RISK_*` | Risk score weights and thresholds (see comments in the file) | various |

### `bounding-boxes.json` — area definitions

Lists the monitoring areas. **The recommended way to manage them is the 🗺️ Areas screen** (which rewrites this file on its own). You can edit it by hand for initial provisioning; if so, **restart** the app.

Each area:

```
"bari": { "name": "Porto di Bari", "keyword": "BARI", "sw": [40.95, 16.60], "ne": [41.30, 1
```

| Field | Meaning |
|---|---|
| key ( `bari` ) | Internal area identifier (also used by `BBOX_PRESET` ) |
| `name` | Name shown in the interface |
| `keyword` | (Optional, can be `null` ) filters “Expected ships” by destination |
| `sw` | South-West corner `[lat, lon]` in decimal degrees |
| `ne` | North-East corner `[lat, lon]` in decimal degrees |

> Manual edits to this file while the app is running only take effect after a restart. If you use the Areas screen, the file’s formatting gets normalized (stays valid, but indentation changes).

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## Backup, restore, and deploy

- **Restoring** a database replaces **all** current data (irreversible): download a backup first. After restoring, data is reassigned to the correct area based on coordinates. Restoring **does not** re-trigger VesselFinder/MarineTraffic scraping (the data is already in the restored DB).
- **Coverage map** data lives in a **separate database**, exportable/importable on its own and still included in the full backup.
- **Auto-restore after a deploy:** the database is wiped whenever the application is updated. If, at startup, the DB **doesn’t exist** and there are **auto-backups** present ( `data/backups/` folder), the app automatically restores the latest backup. Requires the backups folder to survive the deploy. Doesn’t trigger if the DB exists but was simply emptied with “Clear data”. Can be disabled with `AUTO_RESTORE_ON_DEPLOY=false` in `app.config.properties` .

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## Credits

AIS outage detection (the outage banner) relies on the AISStream-Uptime project by buttermilkgreen, an uptime monitor for the AISStream service. The app doesn't embed its code: it only queries the public API of its hosted instance ( `https://aisuptime.buttermilkgreen.fyi` ) to figure out whether a data silence is due to a service outage or simply a quiet area. The project is open source (MIT license) and you can **self-host it**: point `AIS_UPTIME_SELFHOST_URL` at your own instance's URL.